8th April 2026

Virtual Network Service Endpoints

Private Endpoint

Virtual Network Peering

Create & Manage VPN Gateway

Create Site to Site VPN Connection

Gateway Configuration Settings.

Virtual Network Service Endpoints

- **What it is**: Extends your VNet's private IP space to Azure services (like Storage, SQL Database).
- **How it works**: Provides direct connectivity from your VNet to Azure services over Microsoft's backbone, bypassing public internet.
- **Use case**: Securely access Azure services from inside your VNet without exposing traffic to the internet.

Private Endpoint

- **What it is**: A network interface that connects you privately to Azure services using a private IP.
- **How it works**: Traffic flows through your VNet to the service, eliminating exposure to the public internet.
- **Use case**: Access services like Azure Storage or Key Vault securely from within your VNet.

Virtual Network Peering

- **What it is**: Connects two VNets so they act as one for routing purposes.
- **How it works**: Traffic between VNets flows through Microsoft's backbone network with low latency.
- **Use case**: Share resources across VNets (e.g., dev/test and production environments).

Create & Manage VPN Gateway

- **What it is**: A gateway that connects your Azure VNet to other networks (on-premises or other VNets).

- **How it works**: Provides encrypted tunnels using IPsec/IKE protocols.
- **Use case**: Hybrid connectivity between on-premises datacenter and Azure.

Create Site-to-Site VPN Connection

- **What it is**: A VPN tunnel between your on-premises VPN device and Azure VPN Gateway.
- **How it works**: Uses IPsec/IKE to encrypt traffic between the two networks.
- **Use case**: Extend your on-premises network into Azure securely.

Gateway Configuration Settings

- **What it is**: Settings that define how the VPN Gateway operates (SKU, type, IP addresses).
- **How it works**: You choose gateway type (VPN or ExpressRoute), SKU (Basic, VpnGw1–5), and configure IPs.
- **Use case**: Tailor gateway performance and cost to your workload needs.

Step-by-Step in Azure Portal

Here's how you would typically configure these features:

Best Practice:

- Use **Private Endpoints** for maximum security.
- Use **Service Endpoints** when you want simplicity and don't need private IPs.
- Use **VNet Peering** for fast, low-latency communication between VNets.
- Use **VPN Gateway** and **Site-to-Site VPN** for hybrid cloud scenarios.

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Here's a visual overview of the Azure networking features you asked about. These diagrams illustrate how **Service Endpoints, Private Endpoints, VNet Peering, VPN Gateways, Site-to-Site VPNs, and Gateway Configurations** fit together in practice:

- **Virtual Network Service Endpoints** → iturn0image1
- **Private Endpoint Architecture** → iturn0image13
- **Virtual Network Peering** → iturn0image4
- **VPN Gateway Setup** → iturn0image11
- **Site-to-Site VPN Connection** → iturn0image16

- **Gateway Configuration Settings** → iturn0image7

How They Work Together

- **Service Endpoints**: Extend your VNet to Azure services securely over Microsoft's backbone.
- **Private Endpoints**: Provide private IP connectivity to Azure services, eliminating public internet exposure.
- **VNet Peering**: Connects VNets for seamless communication with low latency.
- **VPN Gateway**: Creates encrypted tunnels between Azure and on-premises networks.
- **Site-to-Site VPN**: Extends your on-premises network into Azure securely.
- **Gateway Configurations**: Define gateway type, SKU, and IP settings to match performance and cost needs.

Note:-Together, these features form the backbone of secure hybrid and multi-cloud networking in Azure. Service Endpoints and Private Endpoints secure access to Azure services, VNet Peering connects VNets, and VPN Gateways with Site-to-Site VPNs extend connectivity to on-premises environments.

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Azure Networking Advanced Lab

Scenario

A company wants to:

- Secure access to Azure services privately
- Connect multiple VNets
- Connect on-premises network to Azure

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1. Virtual Network Service Endpoints

Allows **secure connection from VNet → Azure services** (like Storage)

Practical: Enable Service Endpoint for Storage

Step 1: Create VNet

1. Go to **Azure Portal**
2. Create Virtual Network:

    - Name: `VNet-Endpoint`
    - Address: `10.1.0.0/16`
3. Create Subnet:
    - Name: `AppSubnet`
    - Range: `10.1.1.0/24`

Step 2: Enable Service Endpoint

1. Go to **VNet → Subnets**
2. Click `AppSubnet`
3. Under **Service Endpoints**
4. Select:
    - **Microsoft.Storage**

Click **Save**

Step 3: Restrict Storage Account

1. Create Storage Account
2. Go to **Networking**
3. Select:
    - **Selected networks**
4. Add:
    - VNet → `AppSubnet`

Now only this subnet can access storage ✔

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2. Private Endpoint (More Secure)

Provides **private IP inside VNet for Azure service**

Practical: Create Private Endpoint for Storage

Step 1: Create Storage Account

- Name: `mystorageprivate123`

Step 2: Create Private Endpoint

1. Go to **Private Endpoint → Create**
2. Fill:
   - Name: `PE-Storage`
   - VNet: `VNet-Endpoint`
   - Subnet: `AppSubnet`
3. Resource:
   - Type: Storage
   - Select your storage account

Click **Create**

Step 3: Verify

- Storage will now be accessed via **private IP**
- Public access can be disabled ✔

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3. Virtual Network Peering

Connect 2 VNets (like same network)

Practical: Peer Two VNets

Step 1: Create VNets

- `VNet1` → 10.0.0.0/16
- `VNet2` → 10.2.0.0/16

Step 2: Create Peering

1. Go to `VNet1`
2. Click **Peerings → + Add**

Fill:

- Name: `VNet1-to-VNet2`
- Select: `VNet2`
- Allow:

- Virtual network access ✔

Click **Add**

Step 3: Repeat Reverse Peering

Do same from `VNet2 → VNet1`

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4. Create & Manage VPN Gateway

Used to connect **Azure ↔ On-Premises**

Practical: Create VPN Gateway

Step 1: Create Gateway Subnet

1. Go to VNet
2. Add Subnet:
   - Name: `GatewaySubnet`
   - Range: `10.0.255.0/27`

Step 2: Create VPN Gateway

1. Search → **Virtual Network Gateway**
2. Click **Create**

Fill:

- Name: `MyVPNGateway`
- Gateway Type: VPN
- VPN Type: Route-based
- SKU: VpnGw1
- VNet: Select your VNet

Takes ~30–45 minutes

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5. Site-to-Site VPN Connection

Connect Azure VNet to **on-prem network**

Practical Steps

Step 1: Create Local Network Gateway

1. Search → **Local Network Gateway**
2. Create:

- Name: `OnPrem-LNG`
- IP Address: Your router public IP
- Address Space: `192.168.1.0/24`

Step 2: Create Connection

1. Go to **VPN Gateway**
2. Click **Connections** → **+ Add**

Fill:

- Connection Type: Site-to-Site
- Local Network Gateway: `OnPrem-LNG`
- Shared Key (PSK): `abc123`

Click **Create**

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6. Gateway Configuration Settings

Important Settings

Setting	Description
Gateway Type	VPN / ExpressRoute
VPN Type	Route-based (recommended)

Setting	Description
SKU	Performance level
GatewaySubnet	Mandatory subnet
Shared Key	Authentication

Example Configuration

- Gateway Type: VPN
- VPN Type: Route-based
- SKU: VpnGw1
- Address Space:
  - Azure: 10.0.0.0/16
  - On-prem: 192.168.1.0/24

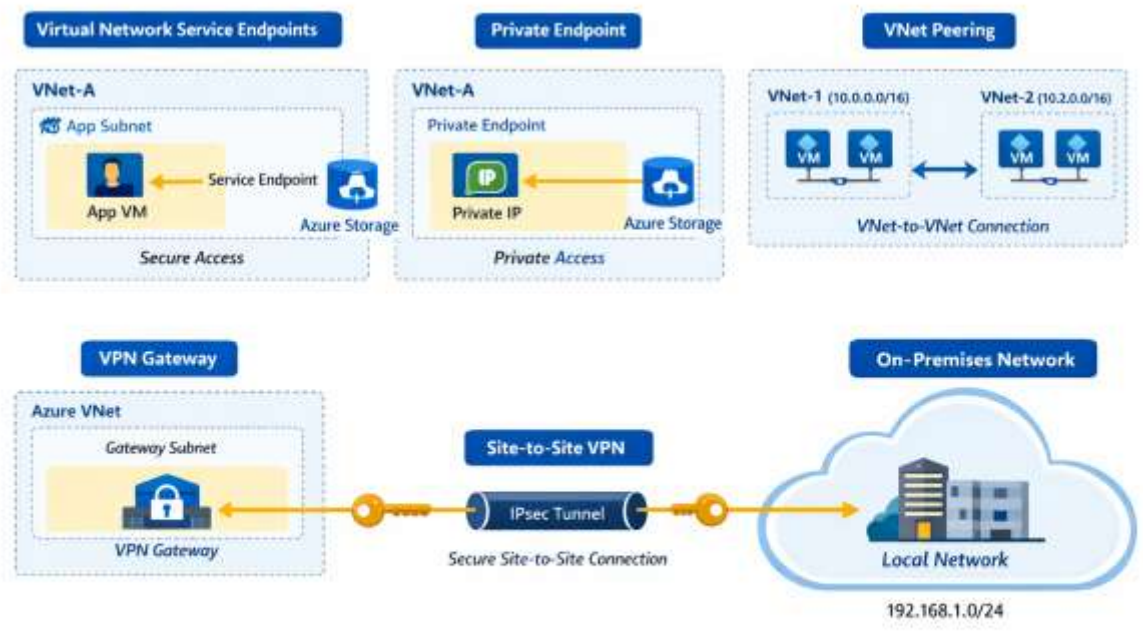
End-to-End Practice Task

Task:

1. Create VNet + Subnet
2. Enable Service Endpoint
3. Create Storage + restrict access
4. Create Private Endpoint
5. Create 2 VNets + Peering
6. Create VPN Gateway
7. Configure Site-to-Site VPN

Interview Questions

1. Difference between Service Endpoint vs Private Endpoint?
2. Why GatewaySubnet is required?
3. Can peered VNets communicate without internet?
4. Route-based vs Policy-based VPN?
5. Difference between Site-to-Site and Point-to-Site?

Best Practices

Use **Private Endpoint** for high security
Use **Peering instead of VPN (within Azure)**
Always use **Route-based VPN**
Plan IP ranges carefully (no overlap)